Binary affine-subspace factor-base input

Preparation record, 2026-10-17 UTC. On the certified ordinary-binary GF(2^61) curve with prime subgroup order r=576,460,752,557,211,763, fix V=span_F_2{1,z,…,z^{10}}. Before target selection, scan four affine offsets 2048, 4096, 6144 and 8192. Keep liftable x with an order-r point, one y representative per x and both signs implicit. Choose realized size nearest the m=8 default B=287, then generate an equal-size random-x control.

Target-blind affine F_2 subspace: realized order-r base size

Arm	Lift fails	Outside r	Accepted x
Offset 2048, selected	995	804	249
Offset 4096	1,025	775	248
Offset 6144	1,028	775	245
Offset 8192	1,024	788	236
Random control	1,099	797	249

The subspace polynomial Q(X) has 12 nonzero Frobenius terms and kernel V. The affine-x prefilter Q(X)+Q(c)=0 has 13 nonzero field terms and 50 independent F_2-linear constraints. Exact membership in this base also requires a curve lift and the order-r subgroup predicate; its solver cost is not measured. Sage independently replayed all four scans, the 2,145 random-control draws, point coordinates, subgroup membership and the predeclared selection rule.

Preparation including rejected offsets and control: 56.143 s wall, 38.774 s CPU, 68,960,256 bytes peak RSS on this shared host. This is descriptive setup cost, not a solver or end-to-end speed comparison. The field's multiplicative group has prime order 2^61−1, so no useful proper multiplicative-coset divisor exists on this exact field.

The next design must price that subgroup predicate or an all-liftable, cofactor-projected base with explicit torsion handling. Input sparsity alone does not establish a lower degree of regularity. Relation yield, rank and target recovery, matched rho, m=2…7 and direct S_8/S_9 comparisons remain unmeasured.

Data and source-bound Sage replay: research/icperf_binary_subspace_base_20261017/.